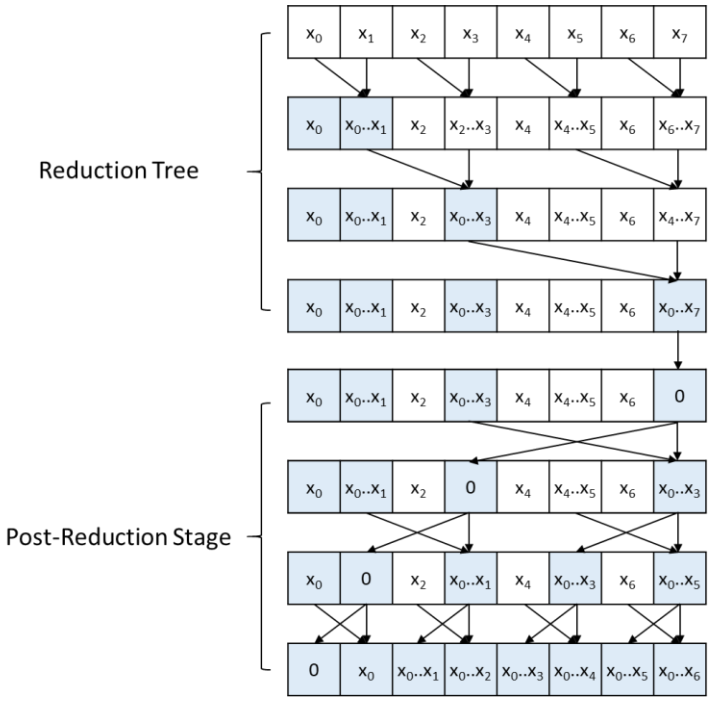
CMPS 224/324: GPU COMPUTING
ASSIGNMENT 7

In this assignment, you will implement an exclusive scan kernel. Instead of using the Kogge-Stone method, you must use another method called the Brent-Kung method. This method is illustrated below:



Your result should be segmented, meaning that each thread block produces a scanned segment. The size of each thread block's segment is 2048 elements. You do not need to consolidate the segments across thread blocks. Your kernel should use shared memory for storing the intermediate results. It should also minimize control divergence by reassigning which elements each thread is responsible for in each iteration such that threads drop out from the end. For example, in the illustration above, in the first iteration, only the first four threads are active, which means that thread 0 is responsible for adding x_0 and x_1. In the second iteration, only the first two threads are active, which means that thread 0 is responsible for adding $x_0..x_1$ and $x_2..x_3$. You do not need to apply warp-level primitives or thread coarsening. Your kernel is expected to work for any set of input dimensions so make sure to handle boundary conditions correctly.

Instructions

1. Place the files provided with this assignment in a single directory. The files are:
   - `main.cu`: contains setup and sequential code
   - `kernel.cu`: where you will implement your code (you should only modify this file)
   - `common.h`: for shared declarations across `main.cu` and `kernel.cu`
   - `timer.h`: to assist with timing
   - `Makefile`: used for compilation

2. Edit `kernel.cu` where TODO is indicated to implement the scan kernel.

3. Compile your code by running: `make`

4. Test your code by running: `./scan`
    - If you are using the HPC cluster, do not forget to use the submission system. Do not run on the head node!
    - For testing on different input sizes, you can provide your own values for input size as follows: `./scan <N>`

Submission

Submit your modified `kernel.cu` file via Moodle by the due date. Do not submit any other files or compressed folders.